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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » The Joy of Computing using Python (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 10 ()

Week 12 : Assignment 12

The due date for submitting this assignment has passed.

Due on 2026-04-15, 23:59 IST.

Assignment submitted on 2026-04-10, 21:50 IST

Understanding PageRank Through Code-Based Simulation

In the lectures, we discussed a Python program that simulates a simplified version of the PageRank algorithm using **point distribution**. The program creates a directed graph, assigns points to nodes, redistributes those points according to outgoing edges, and compares the results with NetworkX's built-in PageRank function.

1. Graph Creation
A directed graph (DiGraph) with 10 nodes (numbered 0 to 9) is created using NetworkX.

2. Random Edge Generation
The function responsible for adding edges loops over all pairs of distinct nodes (s, t) and adds a directed edge from s to t with a **50% probability**. Self-loops are avoided by checking that s != t.

3. Initial Point Assignment
Every node starts with **100 points**. These points are stored in a list where the index corresponds to the node number.

4. Point Distribution Logic
During redistribution:

- If a node has **no outgoing edges**, it keeps all its points.
- If a node has outgoing edges, its points are **split equally** among them.

5. Iterative Redistribution
The program repeatedly redistributes points over multiple rounds. After each round, the new distribution is printed. The user can stop the process by typing #.

Week 11 ()

Week 12 ()

- Page Rank - How does Google Work ? - Part 01 (unit?unit=245&lesson=246)
- Page Rank - How does Google Work ? - Part 02 (unit?unit=245&lesson=247)
- Page Rank - How does Google Work ? - Part 03 (unit?unit=245&lesson=248)
- Page Rank - How does Google Work ? - Part 04 (unit?unit=245&lesson=249)
- Page Rank - How does Google Work ? - Part 05 (unit?unit=245&lesson=250)
- Page Rank - How does Google Work ? - Part 06 (unit?unit=245&lesson=251)
- Page Rank - How does Google Work ? - Part 07 (unit?unit=245&lesson=252)
- Page Rank - How does Google Work ? - Part 08 (unit?

6. Graph Visualization
The graph structure, including nodes and edges, is displayed using matplotlib.

7. Ranking Nodes by Points
At the end, a ranking function stores final points in a dictionary {node: points} and sorts them in ascending order to find which nodes have the lowest and highest scores.

8. Comparison with PageRank
The program also calls NetworkX's built-in pagerank() function to compute the official PageRank values for the same graph and prints them for comparison with the manual simulation.

This simulation helps visualize how **importance flows through a network** and how PageRank emerges as a stable distribution of scores when point redistribution is repeated many times.

1) Why is a **directed graph** mandatory for implementing PageRank in the given code? ***1 point***

- (●) Because PageRank depends on the direction of influence between nodes
- ( ) Because undirected graphs cannot be read from files
- ( ) Because NetworkX only supports directed graphs
- ( ) Because visualization requires arrows

Yes, the answer is correct.
Score: 1

Accepted Answers:
Because PageRank depends on the direction of influence between nodes

2) Why are **all nodes initially assigned equal points** before distribution begins? ***1 point***

- ( ) To speed up execution
- (●) To remove any initial bias in importance
- ( ) To match NetworkX's internal implementation
- ( ) To simplify graph drawing

Yes, the answer is correct.
Score: 1

Accepted Answers:
To remove any initial bias in importance

3) If a node has **many outgoing edges**, how does it affect its contribution? ***1 point***

- ( ) It gains more points
- ( ) It distributes more total points
- (●) Its influence is divided among its neighbors
- ( ) It stops participating after one iteration

Yes, the answer is correct.
Score: 1

Accepted Answers:
Its influence is divided among its neighbors

4) Why does the case study include a **manual Excel-style simulation**? ***1 point***

unit=245&lesson=253)
Page Rank - How does Google Work ? - Part 09 (unit?unit=245&lesson=254)
Page Rank - How does Google Work ? - Part 10 (unit?unit=245&lesson=255)
Page Rank - How does Google Work ? - Part 11 (unit?unit=245&lesson=256)
Page Rank - How does Google Work ? - Part 12 (unit?unit=245&lesson=257)
Page Rank - How does Google Work ? - Part 13 (unit?unit=245&lesson=258)
Page Rank - How does Google Work ? - Part 14 (unit?unit=245&lesson=259)
Page Rank - How does Google Work ? - Part 15 (unit?unit=245&lesson=260)
Page Rank - How does Google Work ? - Part 16 (unit?unit=245&lesson=261)

- To replace Python computation
- To compute final PageRank values
- To reduce graph complexity
- To help visualize a single iteration clearly

Yes, the answer is correct.
Score: 1

Accepted Answers:
To help visualize a single iteration clearly

5) Why is the PageRank process **iterative**? ***1 point***

- Because sorting requires multiple passes
- Because convergence happens gradually
- Because graph size keeps changing
- Because visualization depends on iteration count

Yes, the answer is correct.
Score: 1

Accepted Answers:
Because convergence happens gradually

6) What is the **main purpose** of converting the points list into a dictionary? ***1 point***

- To improve numerical accuracy
- To reduce memory usage
- To speed up sorting
- To preserve node–score association

Yes, the answer is correct.
Score: 1

Accepted Answers:
To preserve node–score association

7) Which type of node can create issues in PageRank distribution? ***1 point***

- Nodes with self-loops
- Nodes with no incoming edges
- Nodes with no outgoing edges
- Nodes with high degree

Yes, the answer is correct.
Score: 1

Accepted Answers:
Nodes with no outgoing edges

8) Why is sorting performed after computing final point values? ***1 point***

- To normalize the graph
- To rank nodes by importance
- To check convergence
- To reduce floating-point error

Collatz Conjecture - Part 01 (unit?unit=245&lesson=262)
Collatz Conjecture - Part 02 (unit?unit=245&lesson=263)
JOC Conclusion (unit?unit=245&lesson=264)
Quiz: Week 12 : Assignment 12 (assessment?name=662)
Week 12 : Programming Assignment 1 (/noc26_cs84/progassignment?name=651)
Week 12 : Programming Assignment 2 (/noc26_cs84/progassignment?name=652)
Week 12 : Programming Assignment 3 (/noc26_cs84/progassignment?name=653)
Week 12 Feedback Form: The Joy of Computing using Python (unit?unit=245&lesson=265)
Text Transcripts ()

Yes, the answer is correct.
Score: 1

Accepted Answers:
To rank nodes by importance

9) What happens if the user **stops the iteration process too early**? ***1 point***

- Results remain accurate
- Values become random
- Convergence still occurs
- Rankings may be misleading

Yes, the answer is correct.
Score: 1

Accepted Answers:
Rankings may be misleading

10) Why is nx.pagerank() used at the end of the case study? ***1 point***

- To replace the manual approach
- To speed up execution
- To visualize the graph
- To validate the conceptual simulation

Yes, the answer is correct.
Score: 1

Accepted Answers:
To validate the conceptual simulation

11) What property of PageRank guarantees reliable results? ***1 point***

- Use of sorting
- Deterministic initialization
- Mathematical convergence
- Absence of cycles

Yes, the answer is correct.
Score: 1

Accepted Answers:
Mathematical convergence

12) Which factor most strongly increases a node's PageRank score? ***1 point***

- Number of outgoing edges
- Random initialization
- Links from already important nodes
- Graph visualization

Yes, the answer is correct.
Score: 1

Accepted Answers:
Links from already important nodes

13) Why is a random graph generated in the later part of the code? ***1 point***

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- To replace real datasets
- To test scalability and behavior
- To guarantee convergence
- To reduce computation time

Yes, the answer is correct.
Score: 1

Accepted Answers:
To test scalability and behavior

14) Why does PageRank still work even if initial points are changed? ***1 point***

- Because initial values are ignored
- Because points are normalized instantly
- Because sorting removes bias
- Because the algorithm converges to the same distribution

Yes, the answer is correct.
Score: 1

Accepted Answers:
Because the algorithm converges to the same distribution

15) Which statement best summarizes the PageRank idea demonstrated in the case study? ***1 point***

- Popular nodes are those with many outgoing edges
- Importance is random but stabilizes
- Every node eventually gets equal rank
- A node is important if important nodes point to it

Yes, the answer is correct.
Score: 1

Accepted Answers:
A node is important if important nodes point to it